Homework 5

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Exercise 1

Part (a)

$$A = \begin{bmatrix} -5 \\ 3 \end{bmatrix}$$

To compute the SVD decomposition of A, we first find the eigenvalues of AA^T.

$$\begin{aligned} AA^T &= \begin{bmatrix} 25 & -15 \\ -15 & 9 \end{bmatrix} \\ \lambda &= 0, 34 \\ \sigma_1 &= \sqrt{34} \end{aligned}$$

The corresponding unit eigenvectors, which are the columns of V, are:

$$\begin{aligned} \begin{bmatrix} 25 & -15 \\ -15 & 9 \end{bmatrix} v_1 &= 34 v_1 \\ v_1 &= \frac{1}{\sqrt{34}} \begin{bmatrix} -5 \\ 3 \end{bmatrix} \\ \begin{bmatrix} 25 & -15 \\ -15 & 9 \end{bmatrix} v_2 &= 0 v_2 \\ v_2 &= \frac{1}{\sqrt{34}} \begin{bmatrix} 3 \\ 5 \end{bmatrix} \end{aligned}$$

The columns of U are then:

$$u_1 = \frac{1}{\sigma_1} A^T v_1 = \begin{bmatrix} 1 \end{bmatrix}$$

The SVD decomposition is then:

$$A = \begin{bmatrix} -5/\sqrt{34} & 3/\sqrt{34} \\ 3/\sqrt{34} & 5/\sqrt{34} \end{bmatrix} \begin{bmatrix} \sqrt{34} \\ 0 \end{bmatrix} \begin{bmatrix} 1 \end{bmatrix}^T$$

Part (b)

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 2 \\ 2 & 1 \end{bmatrix}$$

To compute the SVD decomposition of A, we first find the eigenvalues of AA^T.

$$AA^T = \begin{bmatrix} 5 & 6 & 4 \\ 6 & 8 & 6 \\ 4 & 6 & 5 \end{bmatrix}$$
$$\lambda = 0, 1, 17$$
$$\sigma_1 = \sqrt{17}$$
$$\sigma_2 = 1$$

The corresponding unit eigenvectors, which are the columns of V, are:

$$\begin{bmatrix} 5 & 6 & 4 \\ 6 & 8 & 6 \\ 4 & 6 & 5 \end{bmatrix} v_1 = 17 v_1$$
$$v_1 = \frac{1}{\sqrt{34}} \begin{bmatrix} 3 \\ 4 \\ 3 \end{bmatrix}$$
$$\begin{bmatrix} 5 & 6 & 4 \\ 6 & 8 & 6 \\ 4 & 6 & 5 \end{bmatrix} v_2 = 1 v_2$$
$$v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$
$$\begin{bmatrix} 5 & 6 & 4 \\ 6 & 8 & 6 \\ 4 & 6 & 5 \end{bmatrix} v_3 = 0 v_3$$
$$v_3 = \frac{1}{\sqrt{17}} \begin{bmatrix} 2 \\ -3 \\ 2 \end{bmatrix}$$

The columns of U are then:

$$u_1 = \frac{1}{\sigma_1} A^T v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$
$$u_2 = \frac{1}{\sigma_2} A^T v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

The SVD decomposition is then:

$$A = \begin{bmatrix} 3/\sqrt{34} & -1/\sqrt{2} & 2/\sqrt{17} \\ 4/\sqrt{34} & 0 & 1/\sqrt{2} \\ 2/\sqrt{17} & -3/\sqrt{17} & 2/\sqrt{17} \end{bmatrix} \begin{bmatrix} \sqrt{17} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}^T$$

Part (c)

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 3 & 0 & 0 \end{bmatrix}$$

To compute the SVD decomposition of A, we first find the eigenvalues of AA^T.

$$AA^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\lambda = 1, 2, 3$$

$$\sigma_1 = \sqrt{3}$$

$$\sigma_2 = \sqrt{2}$$

$$\sigma_3 = 1$$

The corresponding unit eigenvectors, which are the columns of V, are:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} v_1 = 3v_1$$

$$v_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} v_2 = 2v_2$$

$$v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} v_3 = 1v_3$$

$$v_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

The columns of U are then:

$$u_1 = \frac{1}{\sigma_1} A^T v_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$u_2 = \frac{1}{\sigma_2} A^T v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$u_3 = \frac{1}{\sigma_3} A^T v_3 = \sqrt{3} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

The SVD decomposition is then:

$$A = \begin{bmatrix} 1/\sqrt{3} & 0 & 0 \\ 0 & 1 & \sqrt{3} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}^T$$

Exercise 2

Let $A = ab^T$ where $a \in \mathbb{R}^m$ and $b \in \mathbb{R}^n$.

Part (a)

We see that:

$$\begin{aligned} AA^T &= ab^T (ab^T)^T \\ &= ab^T ba^T = (b^T b) aa^T \\ &= ||b||^2 aa^T \end{aligned}$$

We see that this matrix has eigenvalue $\sigma_1^2 = ||a||^2 ||b||^2$, with corresponding eigenvector $v_1 = b/||b||_2$. Then $u_1 = \frac{1}{\sigma_1} A v_1 = a/||a||_2$. The largest singular triplet is $(||a||||b||, a/||a||_2, b/||b||_2)$.

Part (b)

Consequently, $||A||_2 = \sigma_1 = ||a||||b||$ and $||A||_F = \sqrt{\mathrm{tr}(AA^T)} = \sqrt{||b||^2 \mathrm{tr}(aa^T)} = \sqrt{||b||^2 ||a||^2} = ||a||||b||$.

Exercise 3

Let A be a symmetric $n \times n$ matrix. It then has the spectral decomposition $A = Q\Lambda Q^T$ where Q is orthogonal and Λ is diagonal. Then:

$$\begin{aligned} AA^T &= (Q\Lambda Q^T)(Q\Lambda Q^T)^T \\ &= Q\Lambda Q^T Q\Lambda Q^T \\ &= Q\Lambda^2 Q^T \end{aligned}$$

The SVD decomposition of A then has $U = Q$, $\Sigma = \Lambda^2$, and $V = Q$.

Exercise 4

Let A be nonsingular, with singular values $\sigma_1 \geq \ldots \geq \sigma_n > 0$ and SVD decomposition $U\Sigma V^T$.

Part (a)

To find the SVD decomposition of A^{-1}:

$$\begin{aligned} A^{-1} &= \left(U\Sigma V^T\right)^{-1} \\ &= \left(V^T\right)^{-1}\Sigma^{-1}U^{-1} \end{aligned}$$

Since U and V are orthogonal:

$$A^{-1} = V^T \Sigma U$$

where $\Sigma^{-1} = \mathrm{diag}(\sigma_1, \ldots, \sigma_n)$.

Part (b)

The singular values of A^{-1} are the inverses of the singular values of A. The left singular vectors of A^{-1} are the right singular vectors of A and vice versa.

Part (c)

Since the 2-norm is equal to the largest singular value, $||A^{-1}||_2 = 1/\sigma_n$.

Part (d)

Since $||A||_2 = \sigma_1$, we see that $\mathrm{cond}(A) = ||A||_2 \cdot ||A^{-1}||_2 = \sigma_1/\sigma_n$.

Exercise 5

Let A be a $n \times n$ matrix with SVD decomposition $U\Sigma V^T$. Then:

$$\begin{aligned} A^T A &= {(U\Sigma V^T)}^T (U\Sigma V^T) \\ &= (V\Sigma U^T)(U\Sigma V^T) \\ &= V\Sigma^2 V^T \end{aligned}$$

The singular values of $A^T A$ are then the squares of the singular values of A. Further, since $A^T A$ is symmetric, the singular values of ${(A^T A)}^{-1}$ are the inverses of the singular values of $A^T A$. Consequently, $\mathrm{cond}(A^T A) = ||A^T A||_2 \cdot ||{(A^T A)}^{-1}||_2 = \sigma_1^2/\sigma_n^2$.

Exercise 6

Let $f(x) = \frac{1}{x} - a$, which has a root at $x = \frac{1}{a}$.

Part (a)

Since $f'(x) = -\frac{1}{x^2}$, the Newton iteration is:

$$\begin{aligned}
x_{k+1} &= x_k - \frac{f(x_k)}{f'(x_k)} \\
&= x_k - \frac{\frac{1}{x_k} - a}{-\frac{1}{x_k^2}} \\
&= x_k + (x_k - ax_k^2) = 2x_k - ax_k^2
\end{aligned}$$

Part (b)

With an initial guess of $x_0 = 0.5$ and $a = 3$, the first four Newton iterations are:

k	x_k
1	$2 \cdot 0.5 - 3 \cdot 0.5^2 = 0.25$
2	$2 \cdot 0.25 - 3 \cdot 0.25^2 = 0.3125$
3	$2 \cdot 0.3125 - 3 \cdot 0.3125^2 \approx 0.3320$
4	$2 \cdot 0.3320 - 3 \cdot 0.3320^2 = 0.3333$

Part (c)

Let $e_k = ax_k - 1$, where x_k is the kth approximation using Newton's method. Then:

$$\begin{aligned}
e_{k+1} &= ax_{k+1} - 1 \\
&= a(2x_k - ax_k^2) - 1 = -(a^2x_k^2 - 2ax_k + 1) = -e_k^2
\end{aligned}$$

Part (d)

For Newton's method to converge, we require that $|e_{k+1}| < |e_k|$. Since $e_{k+1} = -e_k^2$, this implies that $|e_k|^2 < |e_k|$, and so $|e_k| < 1$. This means that the initial guess x_0 must satisfy:

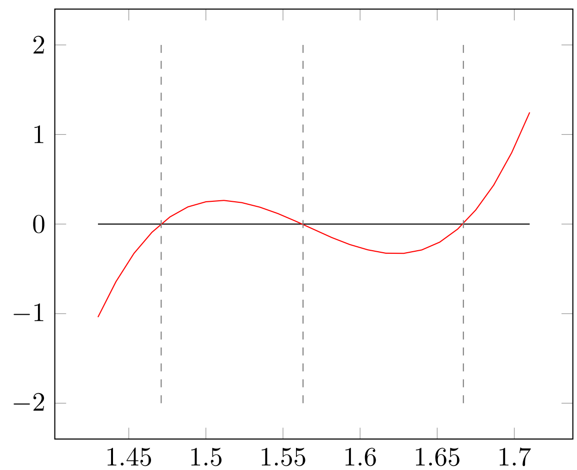
$$\begin{aligned}
|ax_0 - 1| &< 1 \\
-1 < ax_0 - 1 &< 1 \\
0 < ax_0 &< 2 \\
0 < x_0 &< \frac{2}{a}
\end{aligned}$$

Exercise 7

$$p(x) = 816x^3 - 3835x^2 + 6000x - 3125$$

Part (a)

Plotting $p(x)$:



Part (b)

```
EPS = 2.2e-16

def bisection(f, a, b):
    if f(a) * f(b) > 0:
        return None
    i = 0
    c = (a + b) / 2
    while abs(f(c)) > EPS:
        c = (a + b) / 2
        i += 1
        if f(a) * f(c) > 0:
            a = c
        else:
            b = c
    return c, i

p = lambda x : 816* x ** 3- 3835* x ** 2+ 6000* x - 3125
print(bisection(p, 1, 2))
```

Starting with the interval $[a, b] = [1, 2]$, the bisection method converges to the root $x = \frac{25}{17}$ in 41 iterations.

Part (c)

```
EPS = 2.2e-16

def newton(f, f_p, x_0):
    i = 0
    x = x_0 - f(x_0) / f_p(x_0)
    while abs(f(x)) > EPS:
        i += 1
        x = x - f(x) / f_p(x)
    return x, i

p = lambda x : 816* x ** 3- 3835* x ** 2+ 6000* x - 3125
p_ = lambda x : 816* 3* x ** 2- 3835* 2 * x + 6000
print(newton(p, p_, 1.5))
```

Starting with the initial guess $x_0 = 1.5$, Newton's method converges to the root $x = \frac{25}{17}$ in 7 iterations.

Part (d)

```
EPS = 2.2e-16

def secant(f, x_0, x_1):
    i = 0
    x, x_prev = x_1 - f(x_1) * (x_1 - x_0) / (f(x_1) - f(x_0)), x_1
    while abs(f(x)) > EPS:
        i += 1
        x, x_prev = x - f(x) * (x - x_prev) / (f(x) - f(x_prev)), x
    return x, i

p = lambda x : 816* x ** 3- 3835* x ** 2+ 6000* x - 3125
print(secant(p, 1, 2))
```

Starting with the initial guesses $x_0 = 1$ and $x_1 = 2$, the secant method converges to the root $x = \frac{25}{15}$ in 10 iterations.

Exercise 8

$$f(x) = \mathrm{sign}(x-2)\sqrt{|x-2}$$

```python
EPS = 2.2e-16

sign = lambda x : 1if x > 0else -1 if x < 0else 0

def secant(f, x_0, x_1):
    i = 0
    x, x_prev = x_1 - f(x_1) * (x_1 - x_0) / (f(x_1) - f(x_0)), x_1
    while abs(f(x)) > EPS:
        i += 1
        x, x_prev = x - f(x) * (x - x_prev) / (f(x) - f(x_prev)), x
    return x, i

p = lambda x : sign(x - 2) * abs(x - 2) ** 0.5
print(secant(p, 0, 1))
```

Starting with the initial guesses $x_0 = 0$ and $x_1 = 1$, the secant method converges to the root $x = 2$ in 73 iterations.